

Technical Specification & Architectural Whitepaper: Volumetric Optical Processing Unit (VOPU) via Sub-Diffraction Implosion Carving (ImpCarv)

Document ID: SPEC-VOPU-2026-REV6

Classification: Advanced Systems Engineering / Hardware Architecture

Subject: Volumetric Optical Processing Unit (VOPU) with Monolithic 3D Optical Bus

Changelog from REV5: Upgraded architectural paradigm from bulk diffractive volume propagation to a Monolithic 3D Photonic Circuit. Sub-diffraction Implosion Carving (ImpCarv) is now specified to pattern an internal in-volume single-mode optical waveguide network. This internal fiber optic bus routes phase/amplitude-encoded payloads directly to cured metallic/doped terminal nodes (TiO_2 , silver, or gold), eliminating bulk modal dispersion, free-space beam divergence, and K-space degeneracy. Formalized state-adaptive non-linear dynamics including passive ϵ -thresholding via reverse saturable absorbers (RSAs) and continuous λ -decay via excited-state relaxation lifetimes (τ).

1. Executive Summary & Core Paradigm

The Volumetric Optical Processing Unit (VOPU) is a passive, non-silicon hardware architecture that executes high-dimensional Transformer inference via bound volumetric light propagation. By mapping model weight tensors directly into the physical refractive index landscape (Δn) and embedded metallic/doped terminal nodes of a solid-state hydrogel-derived crystal, the VOPU eliminates clock cycles, instruction pipelines, DRAM access, and electronic heat dissipation during compute operations. Computation occurs during the time-of-flight (ToF) of an optical wavefront through the crystal (~ 100 ps across a 3 cm medium), collapsing inference latency to near- $O(1)$ speed-of-light propagation.

Unlike free-space diffractive optical networks that suffer from spatial crosstalk, beam divergence, and K-space degeneracy, REV6 introduces a **Monolithic 3D Photonic Circuit** architecture. Implosion Carving (ImpCarv) patterns a continuous 3D optical waveguide network directly within the hydrogel substrate, routing laser payloads deterministically to cured terminal scattering and reflection nodes.

2. Technology & Fabrication Stack

2.1 Sub-Diffraction Nanofabrication via Implosion Carving (ImpCarv)

The fabrication pipeline utilizes two-photon photopatterning in a swollen, water-rich polyacrylate hydrogel precursor, followed by ionic dehydration to achieve sub-diffraction features:

1. **In-Volume Photopatterning:** A scanned two-photon laser carves micro-channels and defines terminal node coordinates in the expanded hydrogel matrix (~800 nm written resolution).
2. **Terminal Node Functionalization:** Two-photon chemical reduction or cross-linking selectively cures metallic precursors (TiO_2 , silver, or high-index chromophores) at the terminal endpoints of the carved channels.
3. **Isotropic Dehydration Shrinkage:** Ionic dehydration collapses the swollen gel isotropically ($>10\times$ linearly, $\sim 2,000\times$ by volume), compressing the carved channels into single-mode sub-100 nm optical waveguides and contracting the cured terminal nodes into sub-wavelength diffractive and reflection sites.

2.2 Material & Optical Physics Mapping

2.2.1 Monolithic 3D Optical Bus (In-Volume Fiber Network)

The internal optical bus replaces bulk volume scattering with bound single-mode light propagation. Phase- and amplitude-encoded optical payloads launched from an external Spatial Light Modulator (SLM) or laser interferometer enter input waveguide ports at the crystal face. The internal 3D fiber optic channels guide these signals through non-intersecting paths directly to terminal nodes, maintaining mode structure and phase alignment for downstream homodyne sensor detection.

2.2.2 Cured Terminal Nodes & Linear GEMM Zones

Weight matrices ($\mathbf{Wx}$) are executed as incoming guided waves encounter two-photon cured terminal nodes. These nodes act as static beam splitters, phase shifters, and reflective diffractive elements baked directly into the refractive index landscape. Because inputs arrive via bound waveguides rather than unguided fields, interference and matrix operations are strictly deterministic and free from spatial degradation.

2.2.3 Passive State Adaptivity & Lazy Evaluation Mechanics

While weight parameters ($\mathbf{W}$) are permanently frozen in glass, runtime state adaptivity is achieved via passive non-linear material dynamics without electronic control loops:

- **Passive ϵ -Thresholding (Lazy Evaluation):** Zones co-doped with reverse-saturable absorbers (RSAs) remain opaque at low intensities. Sub-threshold activations ($\Delta < \epsilon$) are absorbed, while strong signals bleach the medium and propagate through, executing event-driven sparsity with zero power overhead.

- Continuous λ -Decay (Short-Term Memory):** Excited chromophore states induce temporary refractive index (Δn) and absorption ($\Delta\alpha$) shifts that decay over material relaxation lifetime τ ($n(t) = n_0 \cdot e^{-t/\tau}$). Sequential optical pulses arriving within window τ encounter a modified medium, giving the static crystal dynamic temporal context.
- Dynamic QK^T Attention:** Overlapping Q and K optical fields in dedicated non-linear interaction zones modulate each other's refractive paths in real time via cross-phase modulation, executing dynamic operand multiplication without reconfigurable Mach-Zehnder arrays.

3. End-to-End Compiler & Fabrication Workflow

- Phase 1: Latent Graph Compilation & 3D Topology Routing**
 - 1.1 Extract Transformer state dictionary weights ($\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{W}_{ff}$).
 - 1.2 Synthesize 3D waveguide interconnect topology and terminal node locations.
 - 1.3 Apply inverse shrinkage deformation tensor ($\mathbf{S}^{-1}$) to compensate for $>10\times$ linear contraction.
- Phase 2: Material Synthesis & Two-Photon Scribing**
 - 2.1 Prepare polyacrylate hydrogel matrix doped with photosensitizers, TiO₂/metal precursors, and RSA chromophores.
 - 2.2 Perform two-photon laser scribing of 3D waveguide channels and terminal node coordinates.
- Phase 3: Implosion Carving & Ionic Dehydration**
 - 3.1 Process hydrogel in ionic dehydration bath ($>10\times$ linear / $\sim 2,000\times$ volumetric shrinkage).
 - 3.2 Apply supercritical CO₂ drying to preserve sub-100 nm waveguide geometries.
- Phase 4: Hardware Assembly & Execution**
 - 4.1 Mount monolithic crystal into laser alignment housing.
 - 4.2 Launch phase/amplitude payloads via SLM / interferometry into input optical bus ports.
 - 4.3 Read out speed-of-light transformed output fields at sensor array via homodyne detection.

4. Comprehensive Economics & Cost Comparison

Component / Process	Technical Function	Quantity / Duration	Est. Cost (USD)
Polyacrylate Hydrogel Base	Swellable 3D scaffold precursor	30 mL	\$1.20

TiO ₂ / Metal Precursors	Terminal node reflectors & scattering nodes	<0.5 g	\$0.35
RSA Chromophores	Non-linear gating & ϵ -thresholding	Custom micro-dosing	\$18.50
Photoinitiators & Solvents	Photopatterning & ionic shrinkage bath	1 L bath	\$3.50
Femtosecond Laser Scribing	200 W wall-plug draw (waveguide & node writing)	2 hrs (0.4 kWh)	\$0.08
Dehydration / Drying Energy	Thermal + vacuum supercritical drying	12 hrs (1.5 kWh)	\$0.30
Total Marginal Unit Cost			~\$23.93 / core

5. Primary Literature & Technical References

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